

Solution for Chapter 3

Zimo Guo

3.3

- (a) From left to right, if $\{\mathcal{X}^+ \cup \{\mathbf{x}_{m+1}\}, \mathcal{X}^-\}$ is linearly separable, then the hyperplane is located between the negative samples and the point $\mathbf{x}_{m+1}$. If $\{\mathcal{X}^+, \{\mathbf{x}_{m+1}\} \cup \mathcal{X}^-\}$ is linearly separable, then the hyperplane is located in the positive samples and the point $\mathbf{x}_{m+1}$. So the hyperplane is the line going through origin and $\mathbf{x}_{m+1}$.

From right to left, if $\{\mathcal{X}^+, \mathcal{X}^-\}$ is linearly separable by a hyperplane going through the origin and $\mathbf{x}_{m+1}$, then there exist some hyperplane located between $\mathbf{x}_{m+1}$ and the negative samples and going through the origin can separate $\{\mathcal{X}^+ \cup \{\mathbf{x}_{m+1}\}\}$ and $\{\mathcal{X}^-\}$, and some hyperplane located between $\mathbf{x}_{m+1}$ and the positive samples and going through the origin can separate $\{\mathcal{X}^+\}$ and $\{\{\mathbf{x}_{m+1}\} \cup \mathcal{X}^-\}$. So they are linearly separable.

(b)

$$\begin{aligned}
 \text{Right hand side} &= 2 \sum_{k=0}^{d-1} \binom{m-1}{k} + 2 \sum_{k=0}^{d-2} \binom{m-1}{k} \\
 &= 2 \sum_{k=0}^{d-1} \frac{m-k}{m} \binom{m}{k} + 2 \sum_{k=0}^{d-2} \frac{m-k}{m} \binom{m}{k} \\
 &= 2 \left[\frac{m-0}{m} \binom{m}{0} + \frac{m-1}{m} \binom{m}{1} + \frac{m-2}{m} \binom{m}{2} + \dots + \frac{m-(d-1)}{m} \binom{m}{d-1} \right. \\
 &\quad \left. + \frac{m-0}{m} \binom{m}{0} + \frac{m-1}{m} \binom{m}{1} + \frac{m-2}{m} \binom{m}{2} + \dots + \frac{m-(d-2)}{m} \binom{m}{d-2} \right]
 \end{aligned}$$

Compare with $C(m+1, d)$, we need to prove the following equality,

$$\begin{aligned}
 \frac{d-1}{m} \binom{m}{d-1} &= \sum_{k=0}^{d-2} \frac{m-2k}{m} \binom{m}{k} \\
 \Leftrightarrow (d-1) \binom{m}{d-1} &= (m-2k) \sum_{k=0}^{d-2} \binom{m}{k}
 \end{aligned}$$

- (c) $\Psi(x)$ is the vector of $f(x)$ and is the new input in the linear function. There exists m samples such that every p -subset of $\{\Psi(x_1), \Psi(x_2), \dots, \Psi(x_m)\}$ is linearly independent,

so $\mathbf{a}$ and $\Psi(x)$ play same role with $\mathbf{w}$ and $\mathbf{x}$ in the formula in previous case. Use the equality proved in (b), and obviously $C(m, d) > C(m, d-1)$, so the maximum number of dichotomy is $C(m, p)$. That is $\prod_{\mathcal{F}}(m) = C(m, p) = 2 \sum_{i=0}^{p-1} \binom{m-1}{i}$.

3.4

3.5

$\prod(\mathcal{G}, S)$ is the number of ways for $\mathcal{G}$ to label sample set S . So

$$\prod(\mathcal{G}, S) = |\{(g(x_1), g(x_2), \dots, g(x_m)), g \in \mathcal{G}\}|$$

We want to bound Rademacher complexity in terms of $\prod(\mathcal{G}, S)$, so we need to represent it in terms of the elements in $(g(x_1), g(x_2), \dots, g(x_m))$. Let $(g_j(x_1), g_j(x_2), \dots, g_j(x_m))$ be d_j , and use Massart's lemma,

$$\begin{aligned} \mathfrak{R}_m(\mathcal{G}) &= E_S[E_\sigma[\frac{\sup_{g \in \mathcal{G}} \sum_{i=1}^m \sigma_i g(x_i)}{m}]] \\ &= E_S[E_\sigma[\frac{1}{m} \sup_{d \in \mathbf{b}} \sigma \mathbf{d}]] \\ &\leq E_S[\sqrt{\frac{2 \log \prod(\mathcal{G}, S)}{m}}] \end{aligned}$$

3.6

(a)

$$\begin{aligned} \mathfrak{R}_m(\mathcal{G}) &= E_S[E_\sigma[\frac{\sup_{i=1}^m \sum \sigma_i g(x_i)}{m}]] \\ &= E_S[E_\sigma[\sigma \mathbf{g}]] \\ &= E_S[E_\sigma[\sigma] E_\sigma[\mathbf{g}]] \end{aligned}$$

Because σ is sequence of random variables valued -1 or 1, its expectation is zero vector. So $\mathfrak{R}_m(\mathcal{G}) = 0$ for singleton hypothesis set.

(b)

3.8

(a)

$$\begin{aligned}
\mathfrak{R}_m(\mathcal{H}) &= E_S[E_\sigma[\frac{\sup_{\alpha h \in \alpha \mathcal{H}} \sum_{i=1}^m \sigma_i h(x_i)}{m}]] \\
&= E_S[E_\sigma[\frac{|\alpha| \sup_{\alpha h \in \alpha \mathcal{H}} \sum_{i=1}^m \sigma_i h(x_i)}{m}]] \\
&= E_S[|\alpha| E_\sigma[\frac{\sup_{\alpha h \in \alpha \mathcal{H}} \sum_{i=1}^m \sigma_i h(x_i)}{m}]] \\
&= |\alpha| E_S[E_\sigma[\frac{\sup_{\alpha h \in \alpha \mathcal{H}} \sum_{i=1}^m \sigma_i h(x_i)}{m}]] \\
&= |\alpha| \mathfrak{R}_m(\mathcal{H})
\end{aligned}$$

(b)

$$\begin{aligned}
\mathfrak{R}_m(\mathcal{H} + \mathcal{H}') &= E_S[E_\sigma[\frac{\sup_{h \in \mathcal{H}, h' \in \mathcal{H}'} \sum_{i=1}^m \sigma_i [h(x_i) + h'(x_i)]}{m}]] \\
&= E_S[E_\sigma[\frac{\sup_{h \in \mathcal{H}} \sum_{i=1}^m \sigma_i h(x_i)}{m}]] + E_S[E_\sigma[\frac{\sup_{h' \in \mathcal{H}'} \sum_{i=1}^m \sigma_i h'(x_i)}{m}]] \\
&= \mathfrak{R}_m(\mathcal{H}) + \mathfrak{R}_m(\mathcal{H}')
\end{aligned}$$

(c) If $|f(x_1) - f(x_2)| \leq L|x_1 - x_2|$, $f(x)$ is called **L-Lipschitz function**.To use Talagrand's lemma, we need to prove that $\max(a, b)$ is L-Lipschitz function.

$$\begin{aligned}
|\max(a_1, b_1) - \max(a_2, b_2)| &= \frac{1}{2} |(a_1 - a_2) + (b_1 - b_2) + (|a_1 - b_1| - |a_2 - b_2|)| \\
&\leq \frac{1}{2} |a_1 - a_2| + |b_1 - b_2| + ||a_1 - b_1| - |a_2 - b_2|| \\
&\leq \frac{1}{2} |a_1 - a_2| + |b_1 - b_2| + |(a_1 - b_1) - (a_2 - b_2)| \\
&= \frac{1}{2} |a_1 - a_2| + |b_1 - b_2| + |(a_1 - a_2) - (b_1 - b_2)| \\
&\leq \frac{1}{2} |a_1 - a_2| + |b_1 - b_2| + |a_1 - a_2| + |b_1 - b_2| \\
&= |a_1 - a_2| + |b_1 - b_2| \\
&= \|(a_1, b_1) - (a_2, b_2)\|_1
\end{aligned}$$

So $\max(a, b)$ is 1-Lipschitz function. Then we can use Talagrand's lemma,

$$\mathfrak{R}_m(\max(h, h'))$$

3.9

$h = \max(h_1 + h_2 - 1, 0)$, So

$$\begin{aligned}
\hat{\mathfrak{R}}_S(\mathcal{H}) &= E\left[\sup \frac{\sum \sigma_i h(x_i)}{m}\right] \\
&= E\left[\sup \frac{\sum \sigma_i \max(h_1(x_i) + h_2(x_i) - 1, 0)}{m}\right] \\
&\leq E\left[\sup \frac{\sum \sigma_i (h_1(x_i) + h_2(x_i) - 1)}{m}\right] \\
&= E\left[\sup \frac{\sum \sigma_i h_1(x_i)}{m}\right] + E\left[\sup \frac{\sum \sigma_i h_2(x_i)}{m}\right] + E\left[\sup \frac{\sum \sigma_i}{m}\right] \\
&= \hat{\mathfrak{R}}_S(\mathcal{H}_1) + \hat{\mathfrak{R}}_S(\mathcal{H}_2)
\end{aligned}$$

3.11

(a)

$$\begin{aligned}
\hat{\mathfrak{R}}_S(\mathcal{H}) &= E\left[\sup \frac{\sum_{i=1}^m \sigma_i \sum_{j=1}^{n_2} w_j \sigma(u_j x_i)}{m}\right] \\
&= E\left[\sup \frac{\sum_{j=1}^{n_2} w_j \sum_{i=1}^m \sigma_i \sigma(u_j x_i)}{m}\right] \\
&= \frac{\Lambda'}{m} E\left[\sup \left| \sum_{i=1}^m \sigma_i \sigma(\mathbf{u} x_i) \right| \right]
\end{aligned}$$

(b)

$$\begin{aligned}
\hat{\mathfrak{R}}_S(\mathcal{H}) &= \frac{\Lambda'}{m} E\left[\sup \left| \sum_{i=1}^m \sigma_i \sigma(\mathbf{u} x_i) \right| \right] \\
&\leq \frac{\Lambda'}{m} E\left[\sup \left| \sum_{i=1}^m \sigma_i \mathbf{u} x_i \right| \right] \\
&\leq \frac{\Lambda'}{m} E\left[\sup \sum_{i=1}^m |\sigma_i \mathbf{u} x_i| \right] \\
&= \frac{\Lambda'}{m} E\left[\sup \sum_{i=1}^m |\mathbf{u}| |\sigma_i x_i| \right] \\
&\leq \frac{\Lambda' \Lambda}{m} E\left[\sum_{i=1}^m \sigma_i x_i \right] \\
&= \Lambda' \hat{\mathfrak{R}}_S(\mathcal{H}')
\end{aligned}$$

(c) Cauchy-Schwarz inequality is following,

$$||uv||_2 \leq ||u||_2 ||v||_2$$

Use it in $\hat{\mathfrak{R}}_S(\mathcal{H}')$,

$$\begin{aligned}\hat{\mathfrak{R}}_S(\mathcal{H}') &= \frac{1}{m} E[\sup \sum_{i=1}^m \sigma_1 s \mathbf{u} x_i] \\ &= \frac{1}{m} E[\sup \sum_{i=1}^m \sigma_1 \mathbf{u} x_i] \\ &= \frac{\Lambda}{m} E[|| \sum_{i=1}^m \sigma_i x_i ||_2]\end{aligned}$$

(d)

$$\begin{aligned}\hat{\mathfrak{R}}_S(\mathcal{H}') &= \frac{\Lambda}{m} E[|| \sum_{i=1}^m \sigma_i x_i ||_2] \\ &\leq \frac{\Lambda}{m} \sqrt{E[|| \sum_{i=1}^m \sigma_i x_i ||_2^2]}\end{aligned}$$

(e) Use the previous results and the inequality (3.14) in the book,

$$\begin{aligned}\mathfrak{R}_m(\mathcal{H}) &\leq \hat{\mathfrak{R}}_S(\mathcal{H}) + \sqrt{\frac{\log \frac{2}{\delta}}{2m}} \\ &\leq \frac{\Lambda' \Lambda}{m} \sqrt{E[mr^2]} + \sqrt{\frac{\log \frac{2}{\delta}}{2m}}\end{aligned}$$

3.12

3.14

For a hypothesis set $\mathcal{H}$, the possible most labeling results is $|\mathfrak{H}|$, where every h in the set give a different predict value. So in binary classification tasks, $2^m \leq |\mathcal{H}|$. Get $m \leq \log_2 |\mathcal{H}|$.

In another jiaodu, if VC-dimension of $\mathcal{H}$ is $\log_2 |\mathcal{H}| + 1$, then the hypothesis set should can give $2^{\log_2 |\mathcal{H}|} = 2|\mathcal{H}|$ dichotomy, which is impossible.

3.19

If the dimension of $\mathcal{F}$ is r , consider a set has $m = r + 1$ points, one hypothesis in $\mathcal{H}$ is linear dependent, that is for x_{r+1} , whether it is in this hypothesis could be judged from other x . So if VC-dimension of $\mathcal{H}$ is $r + 1$ and x_{r+1} is linear dependent with x_p and x_q , 2^{r+1} dichotomy is impossible. So $d < r$.

3.20

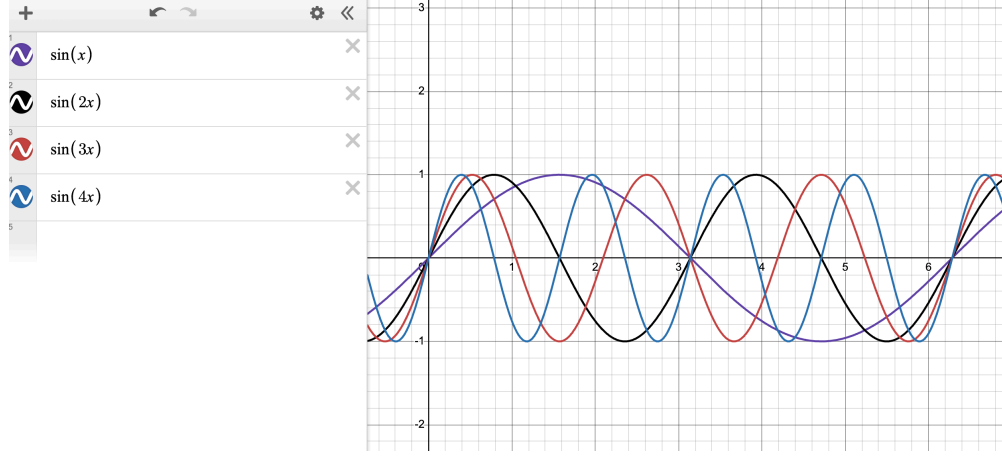


Figure 1: Four Sin Functions in $[0, 2\pi]$

(a) The GCD of the period of four function is 2π . So just consider the interval $[0, 2\pi]$,

From left to right, we can find following 12 dichotomy is possible: $\{+, +, +, +\}$, $\{+, +, +, -\}$, $\{+, +, -, -\}$, $\{+, -, -, +\}$, $\{+, -, +, +\}$, $\{+, -, +, -\}$, $\{-, +, -, +\}$, $\{-, +, -, -\}$, $\{-, +, +, -\}$, $\{-, -, +, +\}$, $\{-, -, -, +\}$, $\{-, -, -, -\}$. There are 4 dichotomy that can not be realized by the sin hypothesis set. So for any x , points $x, 2x, 3x, 4x$ can not be shattered.

(b) doing...

3.27

(a) For a layer, the growth function represent how many possible outputs can be get. For a layer L_2 , $\Pi_{L_1}(m)$ is possible for input, and for a fixed input, output of L_2 has $\Pi_{L_2}(m)$ output, so consider 2 layers, they have $\Pi_{L_1}(m)\Pi_{L_2}(m)$. So for a k layer neuron network,

$$\Pi_{\mathcal{H}}(m) = \Pi_{L_1}(m)\Pi_{L_2}(m) \dots \Pi_{L_k}(m)$$

(b)

3.29

3.30

3.31